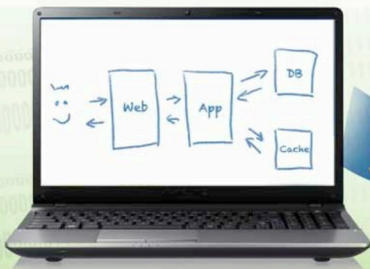


Have You Powered Your Application with Memcached?

Developers know that in computer/network technologies, the speed of an application is really important. Here, we are going to demonstrate how fast a memcached-powered application will perform.



The article published in the March 2012 edition of *OSFY* ('Speed up Your Cloud with Memcached') covered the concepts related to memcached. In this article, I aim to demonstrate just how much performance improves if you use a memcached layer between your application and database.

For example, consider a Web application. Any website that is served up 'dynamically' probably has some static components throughout the life of the page. Loading such information each time through a database query is unacceptable. Let us consider a website that has a very huge list of names in a database, which does not change frequently—and this list has more than 3,000,000 rows of data. Now, you want to display some of the names, which match some user criteria (say, a user clicks on a button). What happens if that page gets reasonably good traffic, and many users click that button to fetch the data? The database will 'cry', right? In this case, every click made by each user will make a direct call to the database, causing a really unacceptable overhead to the database. Here, the importance of caching comes into the picture, and as a caching mechanism, memcached will shine.

Given below are the steps to set up an environment for demonstrating the speed of a memcached-powered application.

Step 1: Setting up the database

Create a database called *ips* in MySQL, and a table called *ips_class* inside it. Then generate a set of 33554431 IPs using bash expansion:

```
echo {192..193}.{0..255}.{0..255}.{0..255}
```

Insert all these IPs into the table *ips_class* using *LOAD DATA INFILE* [see <http://dev.mysql.com/doc/refman/5.5/en/load-data.html>]

```
mysql> select count(*) from ips_class;
+-----+
| count(*) |
+-----+
| 33554431 |
+-----+
1 row in set (18.24 sec)
```